

A ROS 2 Benchmarking Framework for Grid-Based Motion Planning

Comparing BFS, Dijkstra, A*, Theta*, and JPS with Trade-offs in Runtime, Path Quality, and Memory Usage

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Abstract—Motion planning is a fundamental capability for autonomous systems, enabling robots to compute safe and efficient trajectories in environments with obstacles. While the A* algorithm is widely adopted as a baseline, alternative approaches such as Breadth-First Search (BFS), Dijkstra’s algorithm, Theta*, and Jump Point Search (JPS) exhibit different trade-offs in completeness, optimality, path quality, and computational cost. This project proposes a benchmarking framework in the Robot Operating System 2 (ROS 2) that systematically compares BFS, Dijkstra, A*, Theta*, and JPS on two-dimensional occupancy grids. The framework will subscribe to occupancy maps and start/goal poses, compute collision-free paths, and publish results for visualization in RViz. Evaluation will be performed in synthetic environments with varying obstacle densities, grid sizes, and stress-test scenarios such as narrow passages and diagonal corridors. The outcome will be a reproducible ROS 2 package that provides both quantitative metrics (runtime, nodes expanded, path length, number of turns, memory usage) and qualitative visualizations in RViz, highlighting the conditions under which each algorithm is most effective.

I. INTRODUCTION

Robots operating in complex environments must plan paths that are safe, efficient, and computationally feasible. A warehouse robot navigating between shelves or a hospital courier delivering medication through crowded corridors both face the same core challenge: choosing the right motion-planning algorithm for the task at hand. In some cases, an exhaustive method such as Breadth-First Search (BFS) may guarantee completeness but waste time. In others, Dijkstra’s algorithm ensures optimality but explores far more nodes than necessary. A* offers a balance by guiding the search with heuristics, but its effectiveness depends heavily on the environment structure and heuristic design. More advanced grid-based approaches, such as Theta* (which smooths paths through line-of-sight connections) and Jump Point Search (JPS, which accelerates search by pruning unnecessary expansions), highlight additional trade-offs between path quality, runtime, and memory cost.

Despite the prevalence of these algorithms in textbooks and demonstrations, there is limited systematic benchmarking that highlights their relative strengths and weaknesses under controlled conditions. Most examples show a single algorithm on a single map, leaving students and practitioners without clear evidence of when one approach should be favored over another. This project addresses that gap by developing a benchmarking framework in ROS 2 that implements BFS, Dijkstra, A*, Theta*, and JPS in a unified system. The framework will generate synthetic occupancy-grid environments

of varying density and size, execute multiple trials with randomized seeds, and collect performance metrics including runtime, nodes expanded, path length, number of turns, and memory usage. Results will be visualized both in RViz and in performance plots, enabling meaningful comparisons across algorithms.

By moving beyond isolated implementations toward comparative evaluation, this project emphasizes that motion planning is not just about finding a path, but about understanding the trade-offs between completeness, optimality, path quality, efficiency, and scalability.

II. PROJECT TEAM

This project is an individual effort by Mohamed Eljahmi. While group projects are encouraged, all class groups are already formed at this stage. I have therefore expanded the project scope to ensure it remains both substantial and achievable as a solo submission.

The project goes beyond implementing a single planner. It involves developing a modular benchmarking framework in ROS 2, implementing and evaluating multiple algorithms (BFS, Dijkstra, A*, Theta*, and Jump Point Search), designing stress-test scenarios, and producing both quantitative metrics (runtime, nodes expanded, path length, turns, memory usage) and qualitative visualizations in RViz. These tasks together provide sufficient breadth and depth to support a semester-long effort.

All work — including literature review, algorithm development, ROS 2 integration, benchmarking experiments, visualization, analysis, and documentation — will be completed by the proposer.

III. BACKGROUND

Graph-based search methods provide some of the earliest solutions to robotic motion planning. In these approaches, the environment is represented as a discrete graph, with vertices corresponding to feasible configurations and edges representing allowable transitions. Classical algorithms such as Breadth-First Search (BFS) guarantee completeness in finite graphs but are inefficient in terms of computational cost, particularly when applied to large or dense state spaces.

Dijkstra’s algorithm improves upon these methods by assigning weights to edges and expanding nodes according to cumulative path

cost. This ensures optimality of the computed path but requires uniform exploration of the search space, which may result in unnecessary expansions. To address this limitation, Hart, Nilsson, and Raphael introduced the A* algorithm, which augments uniform-cost search with a heuristic function that directs the search toward the goal state. When admissible heuristics are applied, A* maintains optimality while reducing computational effort compared to Dijkstra's method.

Subsequent research has introduced refinements such as **Theta***, which allows line-of-sight connections to produce shorter, more natural paths, and **Jump Point Search (JPS)**, which accelerates grid-based A* by pruning redundant expansions and “jumping” along straight corridors. These algorithms illustrate different trade-offs: Theta* improves path quality at modest computational cost, while JPS dramatically reduces node expansions and runtime in uniform-cost grids.

As emphasized in the course material, these planners illustrate a progression from uninformed search (BFS), to cost-based optimal search (Dijkstra), to heuristic-guided optimal search (A*), and finally to specialized variants (Theta*, JPS) that optimize for particular performance dimensions. While this progression is well known, most demonstrations present a single algorithm in isolation rather than offering a side-by-side evaluation. The lack of systematic benchmarking leaves open questions about when one algorithm should be preferred over another, especially under different map structures, obstacle densities, or heuristic designs.

These graph search techniques have seen widespread adoption across domains such as autonomous driving and warehouse logistics, where efficiency and reliability are critical. Their simplicity, predictability, and guarantees of completeness or optimality make them a strong foundation for motion planning. This project builds on that foundation by developing a benchmarking framework in ROS 2 that implements and compares BFS, Dijkstra, A*, Theta*, and JPS, enabling both **quantitative evaluation** (runtime, nodes expanded, path length, number of turns, memory usage) and **qualitative visualization** (search expansions and paths in RViz) under controlled conditions.

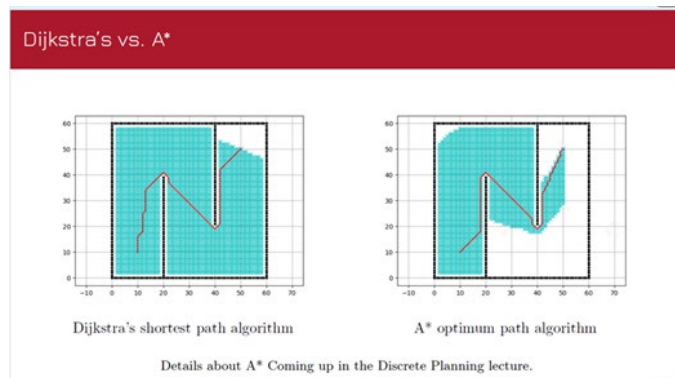


Fig. 1. Comparison of Dijkstra's shortest path algorithm and the A* algorithm. A* incorporates a heuristic function, which reduces the number of expanded nodes while maintaining optimality.

IV. PROPOSED METHODS

The proposed system will be implemented as a **modular motion planning and benchmarking framework** in the Robot Operating System 2 (ROS 2). The framework will operate on graph-based representations of the environment, allowing flexibility in algorithm selection and application domain. In this formulation, vertices correspond to discrete configurations or grid cells, and edges represent feasible transitions between them. The framework will implement five classical and specialized search-based planners:

- **Breadth-First Search (BFS):** provides a baseline for completeness but is computationally inefficient.
- **Dijkstra's Algorithm:** guarantees cost-optimal paths but explores broadly without guidance.
- **A* Search:** balances cost with heuristic guidance (Manhattan or Euclidean distance).
- **Theta* Search:** extends A* by introducing line-of-sight checks, producing shorter and smoother paths.
- **Jump Point Search (JPS):** accelerates A* on uniform-cost grids by pruning redundant neighbors and “jumping” through straight corridors, reducing node expansions.

Each planner will conform to a common interface, enabling interchangeable use within the same ROS 2 node. All planners will maintain open and closed node sets, compute path costs, and reconstruct the final trajectory via backtracking of parent pointers. The ROS 2 package will consist of the following components:

- **Inputs**
 - *Occupancy grid:* Subscribed from the /map topic (nav_msgs/OccupancyGrid). Cells above a specified threshold will be treated as obstacles.
 - *Start and goal poses:* Subscribed from /start and /goal topics (geometry_msgs/PoseStamped).
- **Core algorithms**
 - Implementations of BFS, Dijkstra, A*, Theta*, and JPS.
 - Configurable neighbor connectivity (four- or eight-connected grids).
 - Heuristic functions (Manhattan and Euclidean distance for A* and Theta*).
 - Memory tracking (peak open list size, memory allocations).
- **Outputs**
 - Planned path published on the /planned_path topic (nav_msgs/Path).
 - Logging of performance metrics (runtime, nodes expanded, path length, number of turns, peak memory usage, success/failure).
- **Visualization**
 - Occupancy grid, start/goal poses, and computed paths in RViz.
 - Visualization of node expansions to illustrate algorithmic differences.
 - Side-by-side comparison of paths to highlight trade-offs (shorter vs. smoother vs. faster).
- **Benchmarking**
 - Automated experiments across grid sizes (e.g., 64×64, 128×128) and obstacle densities (10%, 20%, 30%).
 - Stress-test scenarios such as narrow passages, clustered clutter, and diagonal corridors.
 - Multiple randomized seeds per condition to capture performance variability.
 - Generation of summary plots (runtime vs. density, nodes expanded, path length, turns, memory usage) for quantitative comparison.

This modular design ensures that the framework is extensible, allowing future students or researchers to integrate additional planners such as D* Lite, RRT, or PRM with minimal modifications.

V. GOALS AND EXPECTED RESULTS

The primary goal of this project is to develop a **ROS 2 benchmarking framework** that implements and evaluates multiple search-based motion planning algorithms under varied conditions. The emphasis is not only on correctness of implementation but also on generating insights into the trade-offs between completeness, optimality, path quality, computational efficiency, and memory usage.

Specific objectives include:

- Implement **BFS, Dijkstra, A*, Theta*, and JPS** in a unified ROS 2 framework with a common planner interface.
- Support configurable parameters such as neighbor connectivity (four- or eight-connected grids) and heuristic function (Manhattan or Euclidean distance).
- Integrate with ROS 2 message types, subscribing to occupancy grids and start/goal poses, and publishing results as `nav_msgs/Path`.
- Automate benchmarking across multiple grid sizes (e.g., 64×64 , 128×128), obstacle densities (10%, 20%, 30%), and randomized start/goal positions.
- Record and analyze **quantitative metrics**, including:
 - Runtime (ms)
 - Number of nodes expanded
 - Path length
 - Number of turns (for path smoothness)
 - Peak memory usage (open/closed set size)
 - Success rate across trials
- Conduct **stress-test experiments** such as narrow passages, clustered clutter, and diagonal corridors to highlight algorithmic strengths and weaknesses.
- Provide **qualitative outputs** through RViz visualizations of search expansions and paths.
- Deliver reproducible results via CSV logs, analysis scripts, and performance plots.

Expected outcomes:

- A reproducible ROS 2 package enabling side-by-side comparison of BFS, Dijkstra, A*, Theta*, and JPS.
- Benchmarking results that clearly demonstrate trade-offs:
 - BFS/Dijkstra as baselines.
 - A* as heuristic-guided optimal search.
 - Theta* producing smoother and shorter paths.
 - JPS reducing node expansions and runtime in uniform-cost grids.
- Visualizations and figures (runtime curves, expansion plots, path comparisons, memory usage graphs) that make algorithmic differences both measurable and interpretable.
- A flexible foundation that can be extended in future work to include additional algorithms such as D* Lite, PRM, or RRT.

VI. PLANNED SCHEDULE

The project will be completed over the course of the semester in structured phases. Each phase builds on the previous one, ensuring steady progress from algorithm development to benchmarking, analysis, and documentation.

- Weeks 1–2:
 - Literature review of graph search and specialized algorithms (Theta*, JPS, D* Lite).

- Setup of the ROS 2 development environment and package structure.
- Implementation of BFS as the initial baseline planner.
- Weeks 3–4:
 - Implementation of Dijkstra's algorithm within the framework.
 - Validation of BFS and Dijkstra on small synthetic maps.
 - Development of a common planner interface to ensure modularity.
- Weeks 5–6:
 - Implementation of A* with configurable heuristics (Manhattan and Euclidean).
 - Extension to Theta* (line-of-sight path smoothing).
 - Collection of preliminary metrics (runtime, nodes expanded, path length, turns).
- Weeks 7–8:
 - Implementation of Jump Point Search (JPS) with pruning and jump rules.
 - Integration of all planners (BFS, Dijkstra, A*, Theta*, JPS) into a unified ROS 2 node.
 - Initial visualization of node expansions and paths in RViz.
- Weeks 9–10:
 - Benchmarking experiments across grid sizes (64×64 , 128×128) and obstacle densities (10%, 20%, 30%).
 - Stress-test scenarios: narrow passages, clustered clutter, diagonal corridors.
 - Logging of metrics including runtime, nodes expanded, path length, turns, memory usage, and success rate.
- Weeks 11–12:
 - Final analysis of results, highlighting trade-offs between speed, path quality, and memory usage.
 - Generation of plots and figures (runtime curves, expansion plots, memory graphs, RViz screenshots).
 - Stretch goal: demonstration of D* Lite for incremental replanning on dynamic maps.
 - Preparation of final report, presentation, and reproducibility materials (code, CSV logs, analysis scripts).

This schedule ensures that core algorithm implementations are completed by mid-semester, leaving sufficient time for benchmarking, stress tests, analysis, and final deliverables.

VII. CONCLUSION

This project addresses the fundamental problem of motion planning by developing a **benchmarking framework in ROS 2** that goes beyond classical demonstrations of A*. By implementing and systematically comparing BFS, Dijkstra, A*, Theta*, and Jump Point Search (with D* Lite as a stretch goal), the framework will evaluate how different planners trade off completeness, optimality, path quality, runtime efficiency, and memory usage under varied map conditions.

The system will generate reproducible results by combining **quantitative metrics** (runtime, nodes expanded, path length, turns, peak memory usage) with **qualitative outputs** (RViz visualizations

of node expansions and paths). Benchmarking will be conducted across obstacle densities, grid sizes, and stress-test scenarios such as narrow passages and diagonal corridors, making algorithmic strengths and weaknesses clearly visible.

The expected outcome is a comprehensive ROS 2 package that provides a clear picture of when each algorithm is most effective. By moving beyond isolated implementations toward comparative evaluation, this project emphasizes not only how paths are computed, but also what insights can be gained about the trade-offs between speed, path quality, and resource consumption in practical robotics contexts such as warehouse navigation and hospital delivery robots.

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